



Department of Electrical and Electronics Engineering Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. ECE A

Course Code & Title: EC8391 Control Systems Engineering

Name of the Faculty member : Dr.B.Deepa Lakshmi

Innovative Practice Description

- **Unit / Topic:** Unit I / Feed Forward and Feedback Control Theory
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO - 1
- **Activity Chosen:** Class Poll
- **Justification:**

The topic Feed Forward and Feedback Control Theory is the main classification students should know. After teaching the topic, I thought of conducting this activity for making the students to explain the difference between feed forward and feedback control theory which enhance the understanding level and as a subject handler I could judge the learning level of the students.

- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**

After teaching the concept, three minutes has been given to the students to recall about the topic without writing anything. Then class poll starts and I asked the students who could differentiate the given topic. Many students came forward and answer about the difference between the given topics

Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

Students felt that recollecting the topic and answering is good.

❖ *Benefit of the practice:*

1. Most of the students were able to recollect and answer the questions asked.
2. After conducting the activity, I came to know that students are interested to answer.

❖ *Challenges faced in implementation:*

1. Time utilization for conducting activity.



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- Screenshot of the activity:



- CO – PO / PSO mapping:

CO	PO 1	PO 2	PO 3	PO 10	PSO 1
CO1	3	2	2	2	2



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Name of the Faculty member : Dr.B.Deepa Lakshmi

Innovative Practice Description

- **Unit / Topic:** Unit I / Modelling of Mechanical Translational System
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO - 2
- **Activity Chosen:** Finger Signal
- **Justification:**

The topic modeling of mechanical translational system is important to learn and the concept is important to obtain an electrical system. So, after teaching the topic, I thought of conducting this activity for making the students to explain about modeling. So, I pointed out the students and make them answer about modeling which helps me to judge the learning level of the students.

- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**

After teaching the concept, I asked the students one by one to explain about mechanical system. Students enthusiastically answer for this activity. Many students gave the correct answer for the question asked.

Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

Students felt that recollecting the topic and answering is useful for them in remembering the concept.

❖ *Benefit of the practice:*

1. Most of the students were able to recollect and answer for the question.
2. After conducting the activity, I came to know that students have understood the topic and were able to answer.

❖ *Challenges faced in implementation:*

1. Time utilization for conducting activity.



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- Screenshot of the activity:



- CO – PO / PSO mapping:

CO	PO 1	PO 2	PO 3	PO 10	PSO 1
CO1	3	2	2	2	2



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Innovative Practice Description

- **Unit / Topic:** Unit II / PID control
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO - 12
- **Activity Chosen:** Open Book Test in Library
- **Justification:**

The topic is chosen because selection of controller is important for a system to design. After teaching the topic, I planned to conduct this activity for the students to make them understand and write the types of controller. So, I took the students to the library and make them refer the books and write the types of controller and comparing it. This helps me in making the students to learn the concepts and enhance their knowledge.

- **Time Allotted for the Activity:** 30 minutes
- **Details of the Implementation:**

After teaching the concept, I took the students to the library and make them assembled. Then I gave some questions to the students to compare and write the concepts by referring the books available in the library. Students eagerly participated and write the comparisons asked. This enhances the students to analyze the topic and learn quickly.

Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

Students felt that referring more books and writing the concepts gave a wide knowledge about the topic to remember the concept.

❖ *Benefit of the practice:*

1. All the students were able to refer the books and answer for the topic.
2. After conducting the activity, I came to know that students have understood the topic and was able to apply the type of controller for real timer applications.

❖ *Challenges faced in implementation:*

1. Time utilization for conducting activity.



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- Screenshot of the activity:



- CO – PO / PSO mapping:

CO	PO 1	PO 2	PO 5	PO 8	PO 9	PO 10	PSO 1
CO2	3	2	3	1	1	2	2



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Innovative Practice Description

- **Unit / Topic:** Unit III / Frequency Response of standard second order system
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO - 14
- **Activity Chosen:** MATLAB Simulation Tool
- **Justification:**

The topic is chosen because students are studying the frequency response theoretically but there is no lab session for the students to apply and plot the frequency response. So, I took all the students to the lab and make them use individual system to learn how to plot frequency response.

- **Time Allotted for the Activity:** 30 minutes
- **Details of the Implementation:**

After teaching the concept, I took the students to the laboratory and provide a single system for everyone to use MATLAB software. Then I instructed them to obtain frequency response for the given system transfer function. Students eagerly write the coding and simulate it. All the students could able to obtain the response. This enhances the students to know a new simulation tool for analysis.

Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

Students felt that knowing the new tool is interesting and correlating the plot obtained theoretically with the MATLAB output is enhancing their knowledge to compare the concepts.

❖ *Benefit of the practice:*

1. All the students were able to use the MATLAB software.
2. After conducting the activity, I came to know that students could be able to plot the frequency response for any given system.

❖ *Challenges faced in implementation:*

1. Teaching a new tool in half an hour is difficult



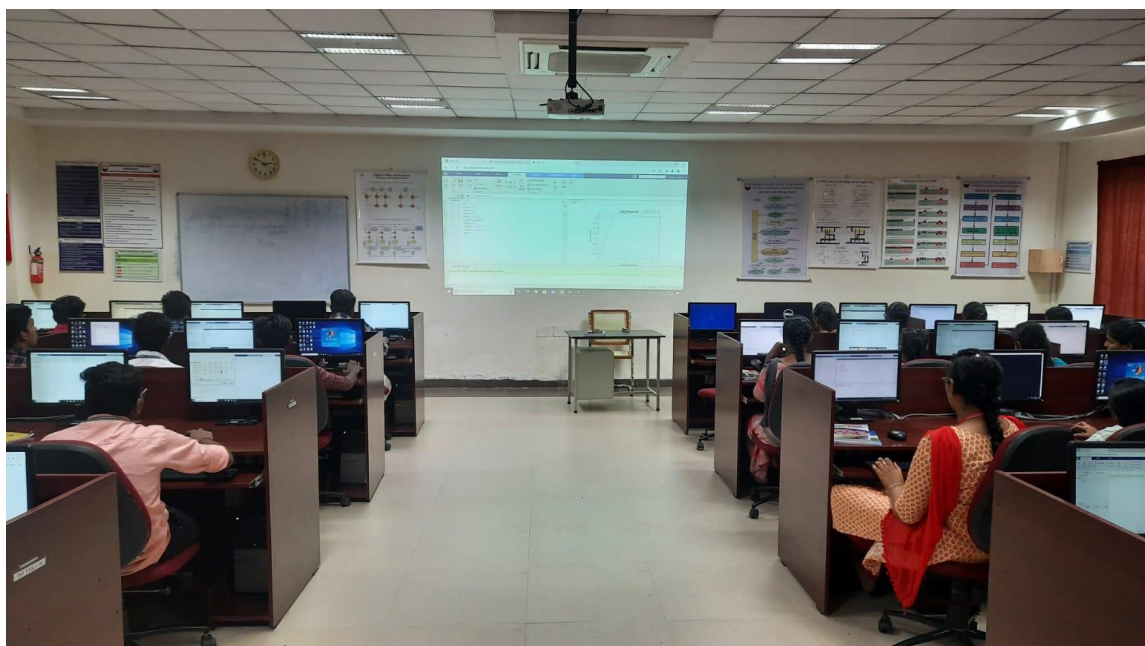
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- Screenshot of the activity:



- CO – PO / PSO mapping:

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 8	PO 9	PO 10	PSO 3
CO3	3	2	3	2	2	1	1	2	2



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Innovative Practice Description

- **Unit / Topic:** Unit IV / Routh Stability Criterion
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO - 18
- **Activity Chosen:** Zero Minute Speech
- **Justification:**

The topic Routh stability Criterion is chosen because the topic is important to determine the stability of the given control system. So, zero minute speech will make the students to recollect the topic and explain about stability analysis. After teaching the topic, I conducted this activity to make the students to understand the topic and explain it.

- **Time Allotted for the Activity:** 10 minutes

- **Details of the Implementation:**

After teaching the concept, I have given a question to the students to analyze and determine the stability of the system. Then I call the students one by one and ask them to explain about the condition for stability and conclusion. Students voluntarily come forward and explain about stability conditions using Routh Stability criterion. The student came to a conclusion after analyzing the given system This enhances the students to analyze and explain the concept in their known way.

- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 5	PO 8	PO 9	PO 10	PSO 1
CO3	3	3	2	2	1	1	1	2

- **PO / PSO mapped:**

Innovative practice	PO 2
	3
Justification for correlation	The student will analyse the stability of the system whether it is stable or unstable and come to a conclusion about stability analysis



- **Screenshot of the activity:**



Reflective Critique:

- ❖ ***Feedback of practice from students and other stakeholders:***

Students felt that explaining about the stability made them understand the concept individually.

- ❖ ***Benefit of the practice:***

1. All the students were able to answer about stability analysis.
2. After conducting the activity, I came to know that students could be able to analyse the stability of the given system using Routh stability criterion.

- ❖ ***Challenges faced in implementation:***

1. Time utilization for conducting the activity

References:

- ✓ J.Nagrath and M.Gopal, "Control Systems Engineering", New Age International Publishers
- ✓ <https://cei.umn.edu/active-learning>



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Innovative Practice Description

- **Unit / Topic:** Unit V / Concepts of controllability and observability
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO - 24
- **Activity Chosen:** Exit Slips
- **Justification:**

The topic controllability and observability is chosen because the topic is important to control the states of the system. So, exit slips will make the students to recollect the topic and write about the concept given. After teaching the topic, I conducted this activity to make the students to understand the topic and write it in a paper.

- **Time Allotted for the Activity:** 10 minutes

- **Details of the Implementation:**

After teaching the concept, I made all the students to write about controllability and observability. Students enthusiastically started writing about the topic and submit. This enhances the students to recollect and write it immediately which will be remembered by them.

- **CO – PO / PSO mapping:**

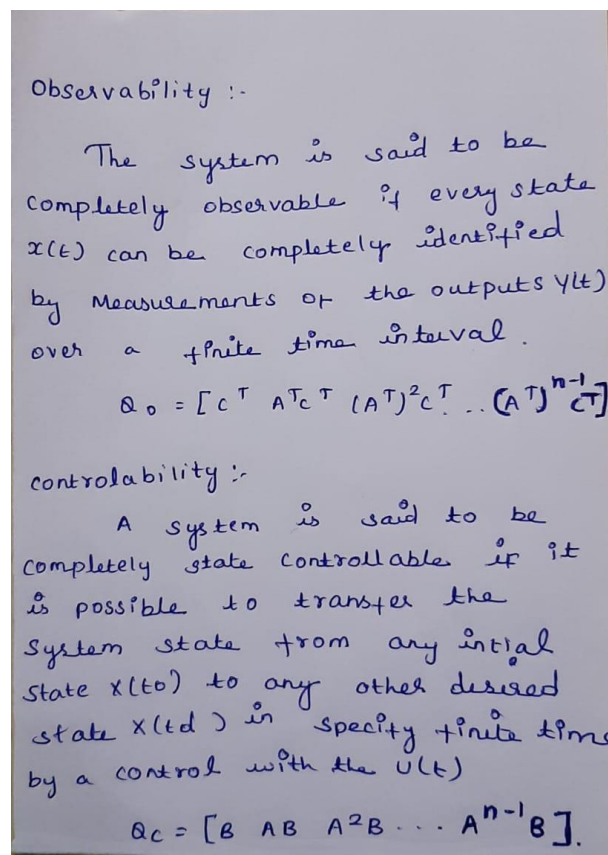
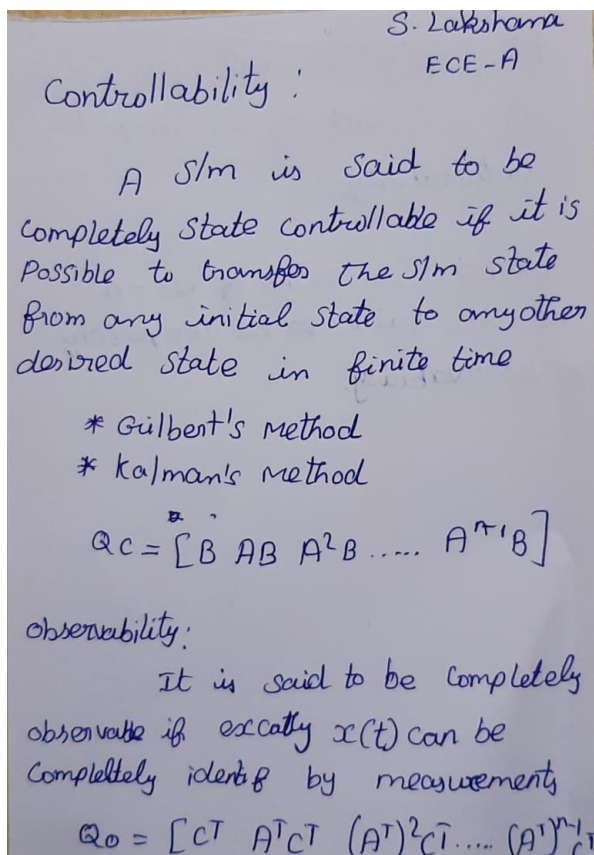
CO	PO 1	PO 2	PO 3	PO 8	PO 9	PO 10	PSO 1
CO5	3	2	2	1	1	1	2

- **PO / PSO mapped:**

Innovative practice	PO 2
	2
Justification for correlation	The student will analyse the controllability and observability of the system to measure states of the system whether the system is state controllable or observable.



• Screenshot of the activity:



Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

Students felt that writing about the given topic make them understand it well and able to recollect immediately.

❖ Benefit of the practice:

1. All the students were able to write the given topic.
2. After conducting the activity, I came to know that students understood the topic and they were able to explain it.

❖ Challenges faced in implementation:

1. Time utilization for conducting the activity
2. Some students could not be able to recollect and write immediately.

References:

- ✓ J.Nagrath and M.Gopal, "Control Systems Engineering", New Age International Publishers
- ✓ https://www.readingrockets.org/strategies/exit_slips



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Innovative Practice Description

- **Unit / Topic:** Unit V / Stability of Linear systems
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO – 25
- **Activity Chosen:** Open Book Test in Library
- **Justification:**

The topic is chosen because analyzing the stability of the given linear system an important concept to learn. After teaching the topic, I planned to conduct this activity for the students to make them understand and analyse the stability of the given system. So, I took the students to the library and make them refer the books and instruct them to analyse the stability for the given system. This helps making the students to learn the concepts and enhance their knowledge.

- **Time Allotted for the Activity:** 30 minutes
- **Details of the Implementation:**

After teaching the concept, I took the students to the library and make them assembled. Then I gave some questions to the students to analyse the concepts by referring the books available in the library. Students eagerly participated and write the analysis asked. This enhances the students to analyze the topic and learn quickly.

Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

Students felt that referring more books and writing the concepts gave a wide knowledge about the topic to remember the concept.

❖ *Benefit of the practice:*

1. All the students were able to refer the books and answer for the topic.
2. After conducting the activity, I came to know that students have understood the topic and was able to analyse the stability for any given system.

❖ *Challenges faced in implementation:*

1. Time utilization for conducting activity.



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CO	PO 1	PO 2	PO 3	PO 8	PO 9	PO 10	PSO 1
CO2	3	2	2	1	1	1	2